

Complexity of Locations and Newsworthiness in a nutshell

More distant events tend to be less interesting. This is explained by the fact that distance adds to cognitive complexity.

In an isotropic 2D space, locations of size $a \times a$ can be numbered by concentric circles around the observer. There are $2\pi d/a$ locations at distance d , and then $\pi d^2/a^2$ locations at a distance smaller than d . The complexity of a given location at distance d amounts to $\log(\pi d^2/a^2)$. It varies like $2 \times \log(d)$.

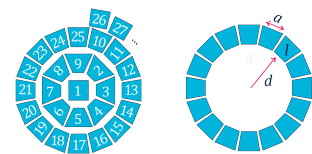
Note that considering polar coordinates will not help. Far locations require much greater accuracy of the angle.

Using landmarks may dramatically reduce complexity, and therefore increase interest.

Minimizing description complexity tells us which landmark to use:

$$L_0 = \operatorname{argmin}(C(L) + 2 \times \log_2(\operatorname{distance}(L, x)))$$

where $C(L)$ can be assessed by $\log_2(r(L))$ when L appears at position $r(L)$ in a list of landmarks.



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